



AINQA
ANCHORING INCLUSIVE HEALTHCARE

MODHS SMILE CDR Implementation Approach

March 2026

MODHS SMILE & KSA National Infrastructure Approach

MODHS Health Ecosystem



Hospitals



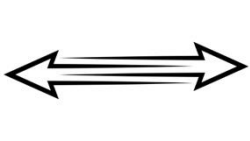
Specialty Digital Service



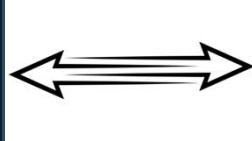
Primary Healthcare Clinics



HL7
CCD
CDA
PFF
CSV



HL7
CCD
CDA
PFF
CSV



MOH Health Ecosystem



E-Prescription



nphies (Unified Health Record)



National Registry



National Vaccination Registry



Anat



Raqeeb



 E-Prescription	A medical prescriptions service that replaces traditional paper prescriptions with an advanced electronic system, making the prescription process more efficient.
 nphies (Unified Health Record)	nphies platform is one of the nation’s pioneering initiatives, led by the Council of Health Insurance in collaboration with the National Health Information Center and under the supervision and support of the Ministry of Health.
 National Registry	Patient verification service that enables stakeholders to electronically verify the data of the patient, healthcare practitioner, or healthcare facility to help serve primary identifier in the healthcare sector.
 National Vaccination Registry	Service that aims to build a unified database containing all vaccinations of individuals, children, and associated information.
 Anat	The Unified Digital Platform for Healthcare Practitioners, registered with the Saudi Commission for Health Specialties.
 Raqeeb	Enables licensed healthcare practitioners to prescribe and dispense controlled drugs while accessing patient medication history to improve decision-making and tracking in healthcare facilities across the Kingdom.

- All of these registries hosted by Lean Business Services
- Connect availability, HL7, FHIR, SOAP, IHE and REST APIs

- NPHIES Release 2 (R2): Clinical Use Cases (HIE)
- Standards: A combination of HL7 v2.7, FHIR R4, IHE XDS, and HL7 CDA R3
- Core Messaging
- ADT: Notifications for Admission, Discharge, and Transfer (HL7 v2.7)
- ORU/ORM: Observation Results and Orders (HL7 v2.7) – Mainly used by Ayenati and Laboratory Systems
- FHIR Clinical Resources
- Medication Request (FHIR R4)
- Medication Dispense (FHIR R4)
- Clinical Documents (through IHE XDS)
- Document Retrieval (Clinical Query & Retrieve)
- IHE Transactions: ITI-18 (Registry Stored Query), ITI-43 (Document Retrieval)
- Content Types: EHR Summary, Immunization Summary (HL7 CDA R3)
- Vitals as well available

Document Type	IHE Transactions	Format
Clinical Documents	ITI-41, ITI-42	HL7 CDA R3
Lab Orders	ITI-41, ITI-42	HL7 CDA R3
Lab Reports	ITI-41, ITI-42	HL7 CDA R3
Radiology Orders	ITI-41, ITI-42	HL7 CDA R3
Radiology Reports	ITI-41, ITI-42	HL7 CDA R3
Radiology Image Manifest	ITI-41, ITI-42, RAD-68	HL7 CDA R3

- Architectural Consulting
- Implementation Approach
- Technical Consulting
- Integration Architecture
- Registry Consolidation